

Dietary inhibitors of monoamine oxidase A.

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Inhibition of monoamine oxidase is one way to treat depression and anxiety. The information now available on the pharmacokinetics of flavonoids and of the components of tobacco prompted an exploration of whether a healthy diet (with or without smoking) provides active compounds in amounts sufficient to partially inhibit monoamine oxidase. A literature search was used to identify dietary monoamine oxidase inhibitors, the levels of these compounds in foods, the pharmacokinetics of the absorption and distribution, and tissue levels observed. An estimated daily intake and the expected tissue concentrations were compared with the measured efficacies of the compounds as inhibitors of monoamine oxidases. Norharman, harman and quercetin dietary presence, pharmacokinetics, and tissue levels were consistent with significant levels reaching neuronal monoamine oxidase from the diet or smoking; 1,2,3,4-tetrahydroisoquinoline, eugenol, 1-piperoylpiperidine, and coumarin were not. Quercetin was equipotent with norharman as a monoamine oxidase A inhibitor and its metabolite, isorhamnetin, also inhibits. Total quercetin was the highest of the compounds in the sample diet. Although bioavailability was variable depending on the source, a healthy diet contains amounts of quercetin that might give sufficient amounts in brain to induce, by monoamine oxidase A inhibition, a small decrease in neurotransmitter breakdown.